

EMBEDDED HACKING

Reverse Engineering the Raspberry Pi Pico 2

Operation Iron Net: Classified Intelligence Briefing

ARM Cortex-M33 • GDB • Ghidra

October 9, 2026

For educational and defensive security research only

```
+-----+
|                TOP SECRET // NOFORN                |
+-----+
| OPERATION IRON NET                                   |
| CLASSIFIED BRIEFING: LIVE CTF 0x03                  |
| NATIONAL SECURITY AGENCY / GMU RHET                  |
+-----+
```

1. The Intercept

At 0200 hours, Forward Operating Base (FOB) Alpha detected a series of covert environmental sensor drops. The adversary is using micro-UAVs to deploy static ground sensors that measure atmospheric conditions (temperature and humidity) via a single-wire protocol. If the conditions match a highly specific, hardcoded threshold, the sensor node transmits a tactical strike authorization over LoRa.

2. The Mandate

CyberCom intercepted one of the sensor nodes. The hardware runs a RP2350. The firmware calculates a complex bitwise checksum (`current_humidity ^ 0xAF & 0x3C`) to validate the environment before authorizing the strike.

Buddy, our AI exploitation model, has failed to crack the checksum logic. You must attach GDB, analyze the bitwise operators in the memory registers, and either manually inject the correct bitmask to force a false-positive strike authorization, or patch the firmware to bypass the bitwise conditional entirely.